

Problem

Two kinds of current-to-voltage converters (also known as *transresistance amplifiers*) are shown in Fig. 5.15.

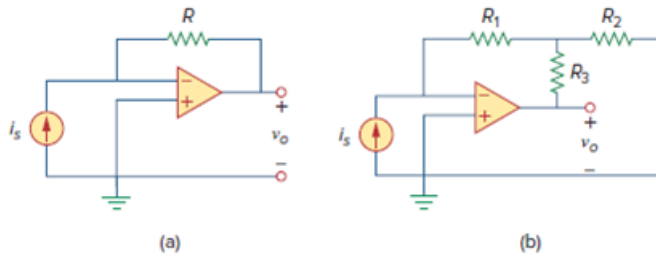
(a) Show that for the converter in Fig. 5.15(a),

$$\frac{v_o}{i_s} = -R$$

(b) Show that for the converter in Fig. 5.15(b),

$$\frac{v_o}{i_s} = -R_1 \left(1 + \frac{R_3}{R_1} + \frac{R_3}{R_2} \right)$$

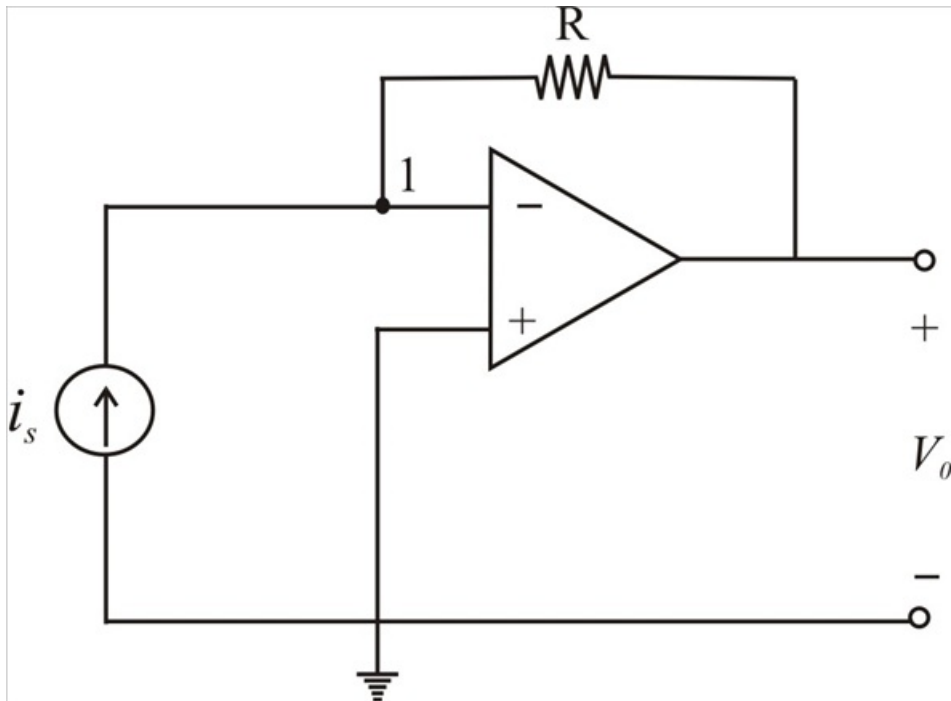
Figure 5.15:



Step-by-step solution

Step 1 of 5

Considering the OP amp circuit



Step 2 of 5

By applying KCL at node (1)

$$\frac{V_1 - V_0}{R} - i_s = 0$$

As the OP amp is ideal

$$V_1 = 0$$

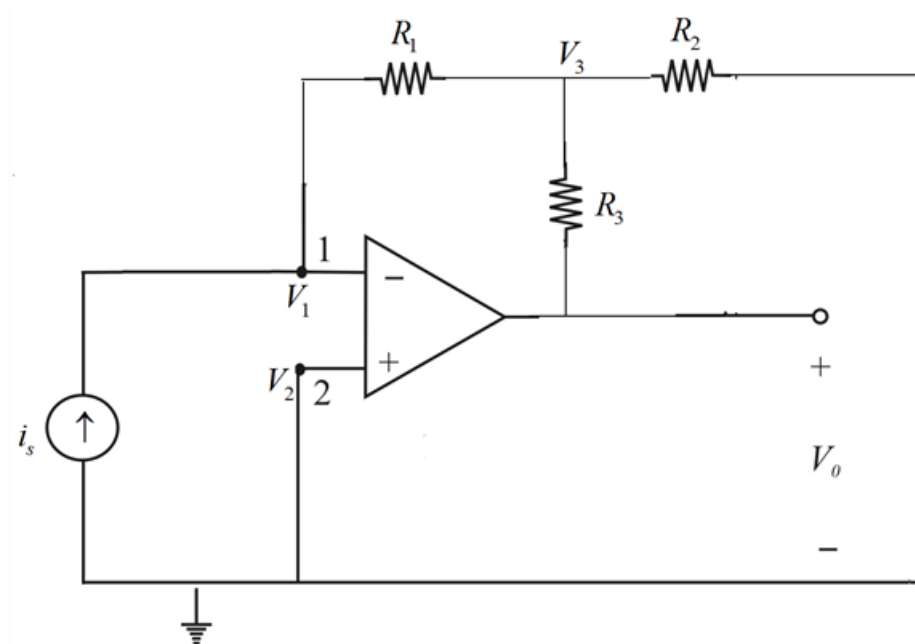
$$\frac{-V_0}{R} - i_s = 0$$

$$\frac{-V_0}{R} = i_s$$

$$\frac{-V_0}{i_s} = R$$

$$\frac{V_0}{i_s} = -R$$

Step 3 of 5



Step 4 of 5

As the OP amp is ideal

$$V_1 = 0$$

$$V_2 = 0$$

By applying KCL at node V_1

$$-i_s + \frac{V_1 - V_3}{R_1} = 0$$

$$i_s = \frac{V_1 - V_3}{R_1}$$

$$i_s = \frac{-V_3}{R_1} \quad (\because V_1 = 0)$$

$$V_3 = -i_s R_1$$

Step 5 of 5

By applying KCL at node V_3

$$\frac{V_3 - V_0}{R_3} + \frac{V_3 - V_1}{R_1} + \frac{V_3}{R_2} = 0$$

$$\frac{V_3 - V_0}{R_3} + \frac{V_3}{R_1} + \frac{V_3}{R_2} = 0 \quad (\because V_1 = 0)$$

$$V_3 \left(\frac{1}{R_3} + \frac{1}{R_1} + \frac{1}{R_2} \right) = \frac{V_0}{R_3}$$

$$V_3 \left(\frac{R_1 R_2 + R_2 R_3 + R_3 R_1}{R_1 R_2 R_3} \right) = \frac{V_0}{R_3}$$

$$V_3 \left(\frac{R_1 R_2 + R_2 R_3 + R_3 R_1}{R_1 R_2} \right) = V_0$$

$$-i_s R_1 \left(\frac{R_1 R_2 + R_2 R_3 + R_3 R_1}{R_1 R_2} \right) = V_0$$

$$\boxed{\frac{V_0}{i_s} = -R_1 \left(1 + \frac{R_3}{R_1} + \frac{R_3}{R_2} \right)}$$